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Quantitative analysis of gut content, Gastro-Somatic Index (GaSI) and Gonado- Somatic Index (GnSI) of the fresh water crab *Acanthopotamon martensi* (Wood-Mason, 1875)

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Abstract- Food is one of the key factors that greatly influence the distribution, growth and reproduction of aquatic or semi-aquatic organisms. Quantitative analysis of Gut content, Gastro-somatic index (GaSI) and Gonado-somatic index (GnSI) of *Acanthopotamon martensi* were studied for a period of one year from the local ponds of Pakur District. The gut content analysis showed that *A. martensi* feeds mainly on diatoms, hydrophytes, organic matter, miscellaneous, chlorophyceae and myxophyceae. Diatoms are the main food items of *Acanthopotamon martensi*. The highest percentage of diatoms were recorded in the summer (50.20 %)and winter season (48.2%). Gastro-somatic index of the male crab was varied from minimum of 0.864 (May) to maximum of 3.651 (January) and the female crab varied from 0.742 (May) to 3.821 (February). The Gonado-somatic index was maximum in the summer season and just after spawning it decreased and came to the minimum value in the winter season. In male as Gonado-somatic index (GnSI) was recorded between 0.089 (January,Month,Temp18°C±0.5) and 0.89 (June, Temp 32.80°C±0.25) whereas in female the Gonado-somatic index between 0.72 (January Temp 18°C ± 0.5) and 8.922 (May temp 29.90°C ± 0.09). There was significant variation ($P<0.05$) in Gonado-somatic index (GnSI) values between two sexes.

Keywords: ***Acanthopotamon martensi*, Gut content, Gastro-somatic index, Gonado- somatic index.**

INTRODUCTION

Food is one of the key factors that greatly influence the distribution, growth, reproduction, migration rate and behavior of organisms.[1] The quality and quantity of food directly affect fish growth while indirectly affect its maturation and survival.[2] Research on food habits and feeding ecology of fishes are the basis to understand their roles within the ecosystem.[3] Both heterotrophs and autotrophs require energy, nutrients and water to survive, grow and reproduce. Lack of food or scarcity of particular nutrients can reduce the growth rate of an organism or even cause failure to reproduce or premature death. In most crustaceans the first second or more pairs of thoracic limbs have become modified for special function and several groups are them as filter. A considerable number of small crabs use the maxillipeds for scooping and collecting detritus. The crabs not go far from home to feed.[4] The food of *Scylla serrata* consists of detritus, crustaceans, molluscs and Polychaetes. Feeding on other crabs is also not uncommon. The females exhibit cannibalistic habit on the mating males inside the burrow.[5]

The analysis of the gut contents of the individuals collected from their habitats should be carried out to know the type of food consumed. Studies on food and feeding

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habit are an important aspect of biology and management of fishes.[6] In general, the feeding intensity varies with seasons, availability of preferred food item, maturity stages and spawning season.[7,8] Studies on the reproductive biology of any species are essential in evaluating the commercial potential of the stocks. Gonado-somatic Index value gives clear cut idea regarding spawning of any species and necessary management measures to promote the fisheries in the concerned localities.[9] Breeding habit of Portunids crabs are interestingly associated with moulting. In portunids soft female mating is characteristic. Copulation takes place just after moult.[10]

Reproduction is one of the prime characters of a living organism through which they perpetuate their race. Crustaceans have complex behavior patterns among these are patterns concerned with reproduction and mating. Mating behavior patterns vary among different species and in number depending upon Physiological condition of the female for their initiation. For successful reproduction, it is essential to bring sexually receptive females and sexually capable males together.

For effective conservation, management and fisheries point of view it is quite essential to study feeding and reproductive behaviour of economically important species of crabs, *Acanthopotamon martensi* (Wood-Mason), a freshwater crab found across the northern Indian subcontinent, including Bangladesh, India, Nepal and Pakistan. In India these species are found in the state of Jharkhand, Bihar, Uttar Pradesh, Uttarakhand and West Bengal. Hence, it requires extensive study on its feeding and reproductive biology of this animal. The present study will mainly deal with quantitative analysis of gut content, gastro-somatic index and gonado-somatic index of the freshwater crab *Acanthopotamon martensi* (Wood-Mason). It will be helpful for biological management of the fisheries of the species.

MATERIAL & METHODS

Live crabs (*Acanthopotamon martensi)* for the present study were collected at regular monthly intervals for one year (January, 2022 to December, 2022) from the local pond of Pakur District and brought to the laboratory for further investigation.

Gut content and Gastro-somatic Index (GaSI)

Soon after the collection, the Crabs were sacrificed, the back was incised and preserved in 5% formalin. After recording the morphometric data of the specimen, the entire gut were dissected out. The stomach was separated, weighed and then reweighed after evacuating the contents into petridish. The difference gave the weight of the gut contents.

The Gastro-somatic Index (GaSI) was calculated by using the formula as suggested.[11]

$$\text{Gastro-somatic Index (GaSI)} = \frac{\text{Weight of the stomach contents}}{\text{Weight of the Crab}} x\ 100$$

The qualitative analysis of the gut was carried out by the volumetric Method (Displacement Method) as suggested.[12] For the plankton analysis of the gut contents Lakey Drop Microtransect Method was adopted.[13]

Gonado-somatic Index (GnSI):

The study the GnSI of *Acanthopotamon martensi* the specimens carried to the laboratory, where they were weighted, narcotized by using ether and the sacrificed to dissect out the gonads. The weight of the gonads of each crab was taken after drying their surface using blotting paper. At the same time, the number of berried crabs was noted in each month. The Gonado-somatic Index of both sexes were calculated for each month using following formula-

$$\text{Gonado-somatic Index (GnSI)} = \frac{\text{Weight of the gonad}}{\text{Weight of the Crab}} x\ 100$$



Fig. 1 - Photograph showing dorsal view of Fresh water crab *Acanthopotamon martensi* (Wood Mason)

RESULTS & DISCUSSION

The analysis of gut content and gastro-somatic Index during the year (January, 2022 to December, 2022) altogether 120 specimens were sacrificed. The results have been presented in the Table -1 and Table-2 for male and female crabs respectively. From these results it is quite clear that *Acanthopotamon martensi* feeds chiefly on

diatoms, hydrophytes, organic matter, miscellaneous, chlorophyceae and myxophyceae in order of preference. However, diatoms were the main food of *Acanthopotamon martensi*. Similar types of result were also recorded in *Paratelphusa spinigera.*[14] The highest amounts of diatoms were recorded in the summer season and the winter season, which might be due to the abundance of these foods in the water and also on the bottom soil. Similar types of summer and winter peak of diatom abundance were recorded.[15]

Gut content analysis plays a crucial role in understanding the feeding habits, food web ecology, trophic level, and dietary requirements of fish species.[16,17] In this study, gut composition of *A. potamon* was found to be diverse and comprised various food items. The analysis revealed the presence of decaying organic matter's in the gut throughout the investigation period. Presence of organic matters confirms that the specimens are detritivores. Higher amount of organic matter were recorded in summer season when the decaying organic matters were abundant in the soli as shown.[18] He was also recorded the similar type of detritus feeding in many porcelain crabs. Due to the extensive tearing and cutting of food materials, miscellaneous food were not identified properly. Probably these materials are of animal origin.

Table 1- Quantitative analysis of Gut content and GaSI value of *Acanthopotamon martensi* (Male) of Local pond of Pakur District of Jharkhand (January, 2022 to December, 2022)

Month	Value of food items in %						GaSI
	Hydrophyte	Bacillarophyceac (Diatoms)	Organic Matter	Miscellaneous	Chlorophyceac	Mixophyceac	
Jan, 2022	-	49.53	31.56	03.65	10.21	05.05	3.651
Feb, 2022	02.12	47.63	22.51	09.70	08.60	09.44	2.630
March, 2022	09.42	42.61	19.46	12.60	05.23	10.68	1.842
April, 2022	08.21	46.82	23.51	17.87	01.61	01.98	1.101
May, 2022	07.20	50.00	26.52	10.34	-	05.76	0.864
June, 2022	04.42	42.60	19.46	15.60	09.30	08.62	0.942
July, 2022	06.25	40.52	29.10	17.15	06.30	00.68	0.902
Aug, 2022	29.40	32.20	22.42	13.36	-	02.62	2.205
Sept, 2022	41.63	36.34	11.58	10.45	-	-	1.705
Oct, 2022	49.44	33.71	08.90	06.43	01.52	-	1.801
Nov, 2022	20.22	40.61	20.10	14.50	-	04.57	1.902
Dec, 2022	11.80	48.00	20.00	07.60	07.26	05.44	2.303

Table 2- Quantitative analysis of Gut content and GaSI value of *Acanthopotamon martensi* (female) of Local pond of Pakur District of Jharkhand (January, 2022 to December, 2022)

Month	Value of food items in %						GaSI
	Hydrophyte	Bacillarophyceac (Diatoms)	Organic Matter	Miscellaneous	Chlorophyceac	Mixophyceac	
Jan, 2022	-	49.43	31.66	03.45	10.31	05.15	3.821
Feb, 2022	02.22	48.53	23.41	08.87	08.63	08.34	2.722
March, 2022	09.52	43.71	120.34	11.44	05.43	09.56	1.862
April, 2022	07.91	47.92	22.46	17.88	01.63	02.20	1.662
May, 2022	07.80	50.40	26.42	10.24	-	05.14	0.742
June, 2022	03.38	42.60	19.86	15.56	08.34	09.26	0.922
July, 2022	06.33	41.52	30.20	13.12	07.21	01.62	0.981
Aug, 2022	29.34	33.25	21.32	13.57	-	02.52	2.202
Sept, 2022	40.63	37.34	12.20	09.83	-	-	1.782
Oct, 2022	48.34	34.81	10.22	04.79	01.84	-	1.802
Nov, 2022	19.82	41.71	19.86	13.69	_	04.92	1.982
Dec, 2022	11.60	48.40	22.24	06.68	07.16	03.92	2.402

Table 3- Mean value (in %) of Gut content analysis of Fresh water crab - *Acanthopotamon martensi* of Local pond of Pakur Distict of Jharkhand (Year- From January, 2022 to December, 2022)

	Hydrophyte	Bacillarophyceac	Organic Matter	Miscellaneous	Chlorophyceac	Mixophyceac
Male	15.843 %	42.548 %	21.26 %	11.604 %	4.161 %	4.57 %
Female	15.574 %	43.385 %	21.682%	10.76 %	4.212 %	4.386 %

Table 4 - Gonado-Somatic Index of Male and Female specimens of *Acanthopotamon martensi* of Local pond of Pakur District, Jharkhand (January, 2022 to December, 2022)

Month	Gonado-Somatic Index (GnSI)	
	Male	Female
Jan, 2022	0.089	0.720
Feb, 2022	0.310	1.380
March, 2022	0.569	2.980
April, 2022	0.610	3.631
May, 2022	0.881	8.922
June, 2022	0.890	8.420
July, 2022	0.768	8.632
Aug, 2022	0.620	4.260
Sept, 2022	0.480	3.190
Oct, 2022	0.260	2.990
Nov, 2022	0.201	2.360
Dec, 2022	0.180	1.820

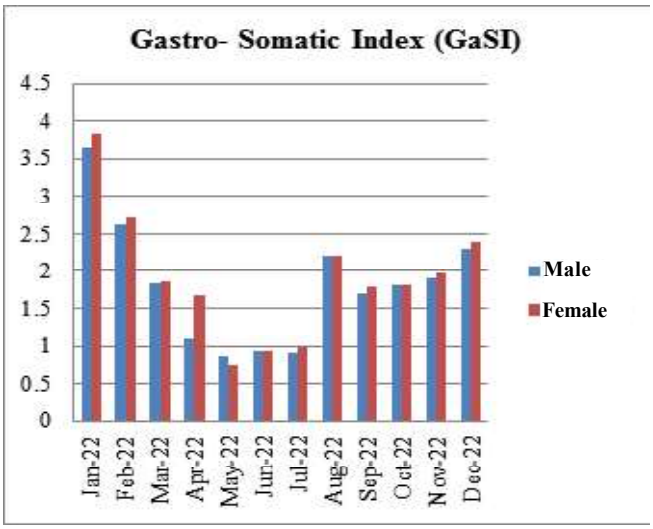


Fig. 1- Monthly variations in mean Gastro-Somatic Index (GaSI) of fresh water crab-*Acanthopotamon martensi* (Wood Mason)

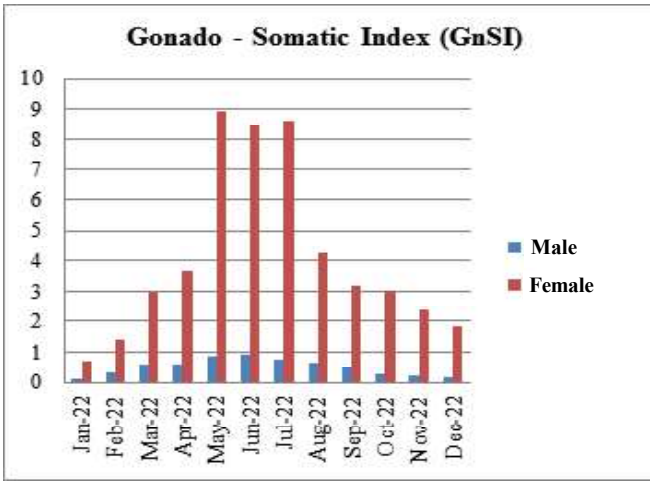


Fig. 2- Monthly variations in mean Gonado- Somatic Index (GnSI) of fresh water crab-*Acanthopotamon martensi* (Wood Mason)

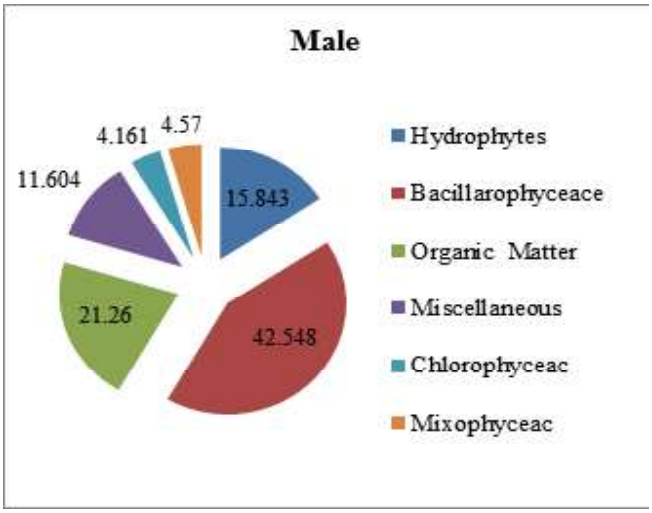


Fig. 3- Pie diagram showing Mean Value (in percentage %) of Gut content analysis of Fresh water crab – *Acanthopotamam martensi* (Male)

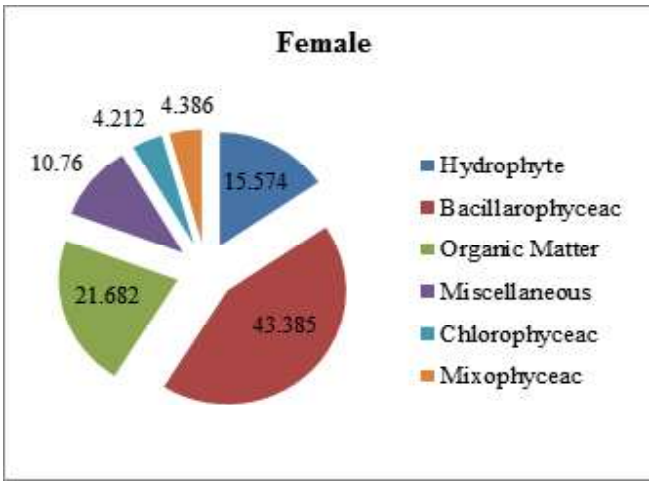


Fig. 4- Pie diagram showing Mean Value (in percentage %) of Gut content analysis of Fresh water crab – *Acanthopotamam martensi* (Female)

It has been observed that the mud crab of Ennore estuary and Pulicate Lake of Tamil Nadu, India subsists mainly on crustaceans followed by mollusks, fishes, detritus, sand and unidentified food materials for both water bodies.[19] Chlorophyceae and myxophyceae were present as few quantity. It reveals that the animal scrap algae with the help of their chelipeds and place them in the mouth for ingestion. Similar type of observation has been recorded.[20]

In the present study, active feeding intensity was observed during the months of November to February with the peak during January (GaSI value 3.651 for male and 3.821 for female) and February (GaSI value 2.630 for male and 2.722 for female), and lowest feeding intensity was

noticed during April to July for both sexes (GaSI value 1.101 to 0.902 for male and 1.662 to 0.981 for female) GaSI value were maximum in prebreeding season. It might be due to the region that in this season animals requires morenutrients for the growth of Gonads. So they take more food for this purpose. Similar type of result was recorded in the case of freshwater crab *Paratelphusa spinigera* by Islam *et al*., (2006)[14]. The higher feeding intensity was observed during prebreeding season and lower feeding intensity was observed in breeding season in case of mud crab *Scylla serata.*[21]

Gonado-somatic Index in *Acanthopotamon martensi* was maximum in the summer season (breeding season) and just after spawning it is decreased and came to the minimum value in the winter season (Table 4). It was observed that freshwater crab breed during the month of May and June. In breeding season the colour of the ovary is bright orange and after this season the colour changes to somewhat pale yellow with inconspicuous in size. Similar type of breeding has also been recorded in different crabs.[22-25]

CONCLUSION

Food is one of the key factors that greatly influence the growth and reproduction of organisms. Gastro-somatic Index (GaS1) indicates feeding intensity, while Gonado-somatic Index (GnSI) indicates reproductive status of the crab *Acanthopotamon martensi*. In case of these fresh water crab *Acanthopotamon martensi* both these Index shows an inverse relationship. If these crab decrease feeding intensity (Lower GaSI) to divert energy to gonad development and increased reproductive behaviour (higher GnSI). The higher Gonado- somatic Index in breeding season indicates the crab is at peak sexual maturity, it will be helpful for the fisheries management.

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